

HORNSBY GIRLS HIGH SCHOOL



Mathematics Extension 1

Year 11

Assessment 3 Term 3 2021

Online

STUDENT NUMBER: _____

General Instructions

- Reading Time – 5 minutes
- Working Time – 40 minutes
- Write using black or blue pen
Black pen is preferred
- NESA-approved calculators and drawing templates may be used
- In Questions 6 – 9, show relevant mathematical reasoning and/or calculations
- Marks may be deducted for untidy and poorly arranged work
- Do not use correction fluid or tape

Total marks – 25

Section I Pages 2 – 3

5 marks

Attempt Questions 1 – 5

Clearly write your answers for Questions 1 – 5 on one sheet of paper for scanning.

You may wish to print off, fill in the objective response answer sheet if and upload a copy of that for section I

Section II Pages 4 – 5

20 marks

Attempt Questions 6 – 9.

Clearly write your answers for Questions 6 – 9 on a separate pages of writing paper for scanning.

Question	1-5	6	7	8	9	Total
Total	/5	/5	/5	/5	/5	/25

This assessment task constitutes 25% of the Preliminary Higher School Certificate Course School Assessment

Section I

5 marks

Attempt Questions 1 – 5

Allow about 10 minutes for this section

Clearly write your answers for Questions 1 – 5 on one sheet of paper for scanning

Q1. If $(x + m)$ is a factor of $8x^3 - 14x^2 - m^2x$ where $m \in R$, $m \neq 0$, then the value of m is

- (A) 7
- (B) 4
- (C) 1
- (D) -2

Q2. Which polynomial has a multiple root at $x = 1$?

- (A) $x^5 - x^4 - x^2 + 1$
- (B) $x^5 - x^4 - x - 1$
- (C) $x^5 - x^3 - x^2 + 1$
- (D) $x^5 - x^3 - x + 1$

Q3. Which of the following is the domain of $f(x) = \sin^{-1}\left(\frac{3}{x}\right)$

- (A) $(-\infty, -3] \cup [3, \infty)$
- (B) $[-3, 3]$
- (C) $\left(-\infty, -\frac{6}{\pi}\right] \cup \left[\frac{6}{\pi}, \infty\right)$
- (D) $\left[-\frac{6}{\pi}, \frac{6}{\pi}\right]$

Q4 The angle θ satisfies $\sin \theta = \frac{12}{13}$ for $\frac{\pi}{2} < \theta < \pi$.

The value of $\cos 2\theta$ is...

(A) $\frac{10}{13}$

(B) $-\frac{10}{13}$

(C) $\frac{119}{169}$

(D) $-\frac{119}{169}$

Q5. Which expression is equivalent to $\frac{\tan 3x - \tan x}{1 + \tan 3x \tan x}$?

(A) $\tan 2x$

(B) $\tan 4x$

(C) $\tan 6x$

(D) $\frac{\tan 3x - 1}{1 + \tan 3x}$

End of Section I

Section II

20 marks

Attempt Questions 6-9

Allow about 30 minutes for this section

Clearly write your answers for Questions 6 – 9
on separate pages for scanning

Question 6 (5 marks) START A NEW PAGE

(a) Find all zeros of $P(x) = x^3 - 6x^2 + 5x + 12$ 3

(b) The roots of $x^3 - 3x + 1 = 0$ are α, β and γ . Find the value of

(i) $\alpha\beta + \alpha\gamma + \beta\gamma$ 1

(ii) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ 1

Question 7 (5 Marks) START A NEW PAGE

(a) Find the exact value of $\sin\left(2\cos^{-1}\left(\frac{2}{3}\right)\right)$ in simplest surd form. 2

(b) Prove that $\tan^{-1}\left(\frac{5}{12}\right) + \cos^{-1}\left(\frac{5}{13}\right) = \frac{\pi}{2}$ 3

Question 8 (5 Marks) START A NEW PAGE

- (a) By letting $t = \tan \frac{\theta}{2}$ or otherwise, show **2**

$$\operatorname{cosec} \theta + \cot \theta = \cot \frac{\theta}{2}$$

- (b) (i) Show $\sin x + \sin 3x = 2 \sin 2x \cos x$ **1**
- (ii) Hence solve $\sin x + \sin 2x + \sin 3x = 0$ for $0 \leq x \leq 2\pi$ **2**

Question 9 (5 marks) START A NEW PAGE

- (a) If the roots of $P(x) = 2x^3 - 7x^2 - 12x + 45$ are α , β and γ , **2**
Find the polynomial whose roots are double that of $P(x)$.
- (b) Solve $x^3 - 5x^2 + 3x + 9 = 0$ given it has a root of multiplicity 2 **3**

End of paper



Mathematics Extension 1

Year 11 Assessment 3 Term 3 2021

STUDENT NUMBER:.....

Section I – Objective Response Answer Sheet

Allow about 10 minutes for this section

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9

A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word **correct** and drawing an arrow as follows.

A ☒ B ☒ ^{correct} C ☐ D ☐

1. A ☐ B ☐ C ☐ D ☐

2. A ☐ B ☐ C ☐ D ☐

3. A ☐ B ☐ C ☐ D ☐

4. A ☐ B ☐ C ☐ D ☐

5. A ☐ B ☐ C ☐ D ☐

Assesment 3 Solutions Ex1 2021

Q1 $8(-m)^3 - 14(-m)^2 - m^2(-m) = 0$

$$-8m - 14m + m^3 = 0$$

$$-7m^2 - 14m^2 = 0$$

$$-7m^2(m+2) = 0$$

$$\therefore m = 0 \quad \text{or} \quad m = -2$$

$$m \neq 0$$

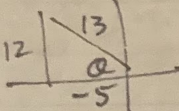
$$\therefore m = -2$$

(D)

Q2 Graph is a shift of a sine half unit to the right ~~left~~ and half as wide. $\therefore 2(x - \frac{1}{2})$

$$\therefore y = \sin^{-1}(2x-1)$$

(B)

Q3. $\sin \theta = \frac{12}{13}$ and Quad 

$$\therefore \cos 2\theta = 2\cos^2 \theta - 1$$

$$= 2 \times \left(-\frac{5}{13}\right)^2 - 1$$

$$= -\frac{119}{169}$$

(D)

Q4 $\frac{\tan 3x - \tan x}{1 + \tan 3x \tan x} = \tan(3x-x) = \tan 2x$ (A)

Q5. $\tan^{-1}(\tan(-\frac{4\pi}{3})) = \tan^{-1}(\frac{\sqrt{3}}{3})$

$$= \tan^{-1}(\tan(-\frac{\pi}{3}))$$

$$= -\frac{\pi}{3}$$

(B)

Question 6

$$(a) \quad p(x) = x^3 - 6x^2 + 5x + 12$$

$$p(1) = 1 - 6 + 5 + 12 \neq 0$$

$$p(-1) = -1 - 6 - 5 + 12 = 0$$

$\therefore (x+1)$ is a factor

$$\begin{array}{r} x^2 - 7x + 12 \\ x+1 \overline{) x^3 - 6x^2 + 5x + 12} \\ \underline{x^3 + x^2} \\ -7x^2 + 5x \\ \underline{-7x^2 - 7x} \\ 12x + 12 \\ \underline{12x + 12} \\ 0 \end{array}$$

$$\begin{aligned} \therefore p(x) &= x^3 - 6x^2 + 5x + 12 \\ &= (x+1)(x^2 - 7x + 12) \\ &= (x+1)(x-3)(x-4) \\ \therefore \text{zeros are } -1, 3, 4 \end{aligned}$$

25% of students did not show the substitution of $P(-1)$ to obtain zero and fails to communicate what it implies

A few students did not pay attention to the difference between a factor and a zero.

Majority did well showing, and communicating what they are doing.

$$(b) \quad x^3 - 3x + 1 = 0$$

$$\alpha + \beta + \gamma = 0$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{-3}{1}$$

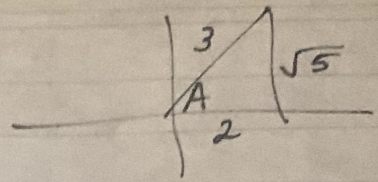
$$\alpha\beta\gamma = \frac{-1}{1}$$

$$(i) \quad \alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a} = \frac{-3}{1} = -3 \quad \text{Many did these well.}$$

$$\begin{aligned} (ii) \quad \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} &= \frac{\alpha\beta + \alpha\gamma + \beta\gamma}{\alpha\beta\gamma} \\ &= \frac{-3}{-1} \\ &= 3 \end{aligned}$$

Question 7(a) $\sin\left(2\cos^{-1}\left(\frac{2}{3}\right)\right)$

$$\begin{aligned} &= \sin 2A \\ &= 2\sin A \cos A \\ &= 2 \times \frac{\sqrt{5}}{3} \times \frac{2}{3} \\ &= \frac{4\sqrt{5}}{9} \end{aligned}$$



Generally well done.
Some students not
able to use double
angle formulae.

(b) RTP $\tan^{-1}\left(\frac{5}{12}\right) + \cos^{-1}\left(\frac{5}{13}\right) = \frac{\pi}{2}$

ie RTP say $\cos\left(\tan^{-1}\left(\frac{5}{12}\right) + \cos^{-1}\left(\frac{5}{13}\right)\right) = \cos \frac{\pi}{2}$

$$\text{LHS} = \cos\left(\tan^{-1}\frac{5}{12} + \cos^{-1}\frac{5}{13}\right)$$

$$= \cos(A+B) \quad \text{where } A = \tan^{-1}\frac{5}{12}$$

$$= \cos A \cos B - \sin A \sin B$$

$$= \frac{12}{13} \cdot \frac{5}{13} - \frac{5}{13} \cdot \frac{12}{13}$$

$$= 0$$

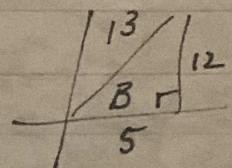
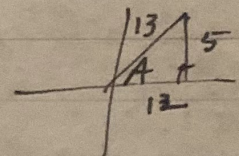
$$\text{RHS} = \cos \frac{\pi}{2}$$

$$= 0$$

$$\therefore \text{LHS} = \text{RHS}$$

and
where

$$B = \cos^{-1}\frac{5}{13}$$



$$\therefore \tan^{-1}\left(\frac{5}{12}\right) + \cos^{-1}\left(\frac{5}{13}\right) = \frac{\pi}{2}$$

Generally well done. Many other solutions are also acceptable,
eg. using Pythagoras or $\sin^{-1}\frac{5}{13} + \cos^{-1}\frac{5}{13} = \frac{\pi}{2}$.

Q 8 a) By letting $t = \tan \frac{\theta}{2}$ or otherwise, show $\operatorname{cosec} \theta + \cot \theta = \cot \frac{\theta}{2}$

$$\text{LHS} = \operatorname{cosec} \theta + \cot \theta$$

$$= \frac{1}{\sin \theta} + \frac{1}{\tan \theta}$$

$$= \frac{1+t^2}{2t} + \frac{1-t^2}{2t}$$

$$= \frac{1+t^2+1-t^2}{2t}$$

$$= \frac{2}{2t}$$

$$= \frac{1}{t}$$

$$= \frac{1}{\tan \frac{\theta}{2}}$$

$$= \cot \frac{\theta}{2}$$

$$= \text{RHS}$$

1 for correct substitution of t results

1 for showing the required result

b) (i) show $\sin x + \sin 3x = 2 \sin 2x \cos x$

$$\text{LHS} = \sin x + \sin 3x$$

$$= \sin 3x + \sin x$$

$$= 2 \sin \left(\frac{3x+x}{2} \right) \cos \left(\frac{3x-x}{2} \right) \quad 1 \text{ mark}$$

$$= 2 \sin \frac{4x}{2} \cos \frac{2x}{2}$$

$$= 2 \sin 2x \cos x$$

$$\text{LHS} = \text{RHS}$$

students need to ensure that they give sufficient working to demonstrate understanding to the marker.

$$\text{or } \text{LHS} = \sin x + \sin 3x$$

$$= \sin x + \sin (2x+x)$$

$$= \sin x + \sin 2x \cos x + \cos 2x \sin x$$

$$= \sin x (1 + \cos 2x) + 2 \sin x \cos x \cdot \cos x$$

$$= \sin x (1 + 2 \cos^2 x - 1) + 2 \sin x \cos^2 x$$

$$= 2 \sin x \cos^2 x + 2 \sin x \cos^2 x$$

$$= 4 \sin x \cos^2 x$$

$$= 2 \times 2 \sin x \cos x \times \cos x$$

$$= 2 \sin 2x \cos x$$

$$\text{LHS} = \text{RHS}$$

$$\text{or } \text{LHS} = \sin x + \sin 3x$$

$$= \sin 3x + \sin x$$

$$= \sin (2x+x) + \sin (2x-x) \quad \text{using } \sin (A+B) + \sin (A-B)$$

$$= 2 \sin 2x \cos x \quad = 2 \sin A \cos B$$

(ii) Hence solve $\sin x + \sin 2x + \sin 3x = 0$ for $0 \leq x \leq 2\pi$

$$\sin x + \sin 2x + \sin 3x = 0$$

$$(\sin x + \sin 3x) + \sin 2x = 0$$

$$2 \sin 2x \cos x + \sin 2x = 0 \quad \text{from part i}$$

$$\sin 2x (2 \cos x + 1) = 0$$

$$\sin 2x = 0$$

$$\text{since } 0 \leq x \leq 2\pi$$

$$0 \leq 2x \leq 4\pi$$

$$2x = 0, \pi, 2\pi, 3\pi, 4\pi$$

$$x = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi$$

Some students missed the last two solutions

$$2 \cos x + 1 = 0$$

$$2 \cos x = -1$$

$$\cos x = -\frac{1}{2}$$

$$\frac{\sqrt{}}{} \frac{\pi}{} \frac{c}{}$$

$$x = \pi - \frac{\pi}{3}, \pi + \frac{\pi}{3}$$

$$x = \frac{2\pi}{3}, \frac{4\pi}{3}$$

Some students gave an additional two solutions

$$x = 0, \frac{\pi}{2}, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}, \frac{3\pi}{2}, 2\pi$$

Question 9 (a)

(a) If $P(x) = 2x^2 - 7x - 12x + 45$
then a polynomial with roots
double of this:

* Method 3 (Longest method)
Students factored
original. Doubled all its
roots then rewrote the required
polynomial in factored
form $y = (x-6)^2(x+5)$

Method 1 let say $X = 2x \therefore x = \frac{X}{2}$

$$\therefore P(x) = 2\left(\frac{X}{2}\right)^2 - 7\left(\frac{X}{2}\right) - 12\left(\frac{X}{2}\right) + 45$$

$$= \frac{X^2}{2} - \frac{7X}{2} - 6X + 45$$

$\therefore$ a polynomial that has the roots
double of $P(x)$ is any of the form

$$P(X) = X^3 - 7X^2 - 24X + 180$$

If students used
this method it
was done well

Although a few
put in $2x$ rather
than $\frac{x}{2}$

Some did x^2 then $5x$

Method 2 If $P(x) = 2x^2 - 7x - 12x + 45$

$$\alpha + \beta + \gamma = \frac{7}{2}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = -\frac{18}{2} = -9$$

$$\alpha\beta\gamma = -45$$

$$\therefore 2\alpha + 2\beta + 2\gamma = 7$$

$$2\alpha \cdot 2\beta + 2\alpha \cdot 2\gamma + 2\beta \cdot 2\gamma = -24$$

$$\text{or } 4(\alpha\beta + \alpha\gamma + \beta\gamma) = -24$$

$$2\alpha \cdot 2\beta \cdot 2\gamma = 8(\alpha\beta\gamma) = -180$$

$$\therefore P(x) = x^3 - 7x^2 - 24x + 180$$

Mixed results using this
method.

A lot of students
only doubled roots 2
at a time and 4 at
a time

Two at a time is $\times 4$
NOT $\times 2$

Three at a time is $\times 8$
NOT $\times 2$

Question 9

(a) If $P(x) = 2x^3 - 7x^2 - 12x + 45$
then a polynomial with roots
double of this:

Method 1 Let say $X = 2x \therefore x = \frac{X}{2}$

$$\therefore P(x) = 2\left(\frac{X}{2}\right)^3 - 7\left(\frac{X}{2}\right)^2 - 12\left(\frac{X}{2}\right) + 45$$

$$= \frac{X^3}{4} - \frac{7X^2}{4} - 6X + 45$$

$\therefore$ a Polynomial that has the roots
double of $P(x)$ is any of the form

$$P(X) = X^3 - 7X^2 - 24X + 180$$

Method 2 If $P(x) = 2x^3 - 7x^2 - 12x + 45$

$$\alpha + \beta + \gamma = \frac{7}{2}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = -\frac{24}{2}$$

$$\alpha\beta\gamma = -45$$

$$\therefore 2\alpha + 2\beta + 2\gamma = 7$$

$$2\alpha \cdot 2\beta + 2\alpha \cdot 2\gamma + 2\beta \cdot 2\gamma = -24$$

$$\text{or } 4(\alpha\beta + \alpha\gamma + \beta\gamma) = -24$$

$$2\alpha \cdot 2\beta \cdot 2\gamma = 8(\alpha\beta\gamma) = 180$$

$$\therefore P(x) = x^3 - 7x^2 - 24x + 180$$

* Method 3 (Longest method)
students factorised
original. Doubled all its
roots then rewrote the required
polynomial in factored
form $y = (x-6)^2(x+5)$

If students used
this method it
was done well

Although a few
put in $2x$ rather
than $\frac{x}{2}$

Some did x^2 then $\sqrt{x}$

Mixed results using this
method.

A lot of students
only doubled roots 2
at a time and 4 at
a time

Two at a time is $\times 4$
NOT $\times 2$

Three at a time is $\times 8$
NOT $\times 2$